# Linear Momentum Problems

Answer key

**Conventions used throughout**

• **Symbols (from the lessons):** $\vec{p} = m\vec{v}$ is momentum (kg·m/s). A prime (′) means *after* the collision; no prime means before. Subscripts name the object.

• **Method (from the lesson):** draw a before and after diagram, choose the + direction, then write $\vec{p}_{tot} = \vec{p}_{tot}'$, that is $m_1v_1 + m_2v_2 = m_1v_1' + m_2v_2'$.

• **Signs:** in one dimension every velocity carries its sign in the working ($v = -3.0$ m/s). The direction is written in square brackets only in the final answer.

• **Elastic head-on collisions:** the key uses the lesson's two equations (Lesson 11), $v_1' = ((m_1 - m_2)/(m_1 + m_2))v_1 + (2m_2/(m_1 + m_2))v_2$ and the matching one for $v_2'$.

• **Two dimensions:** the x-momentum and the y-momentum are conserved separately. Final directions are written like [S 38° E]: start at south and turn 38° toward east.

• **Final answers:** a velocity on a chosen axis is given as the scalar with its sign, then as a vector with its direction in square brackets.

• **Solve in letters first:** each unknown is isolated as a formula before any numbers go in.

• $g = 9.81$ m/s$^2$. Every step keeps full precision; only final answers are rounded.

QUESTION 1 [E]

## Railway cars couple

1. [E] A loaded railway car of mass 6000 kg is rolling to the right at 2.0 m/s when it collides and couples with an empty freight car of mass 3000 kg, rolling to the left on the same pair of tracks at 3.0 m/s. What is the speed and direction of the pair after the collision?

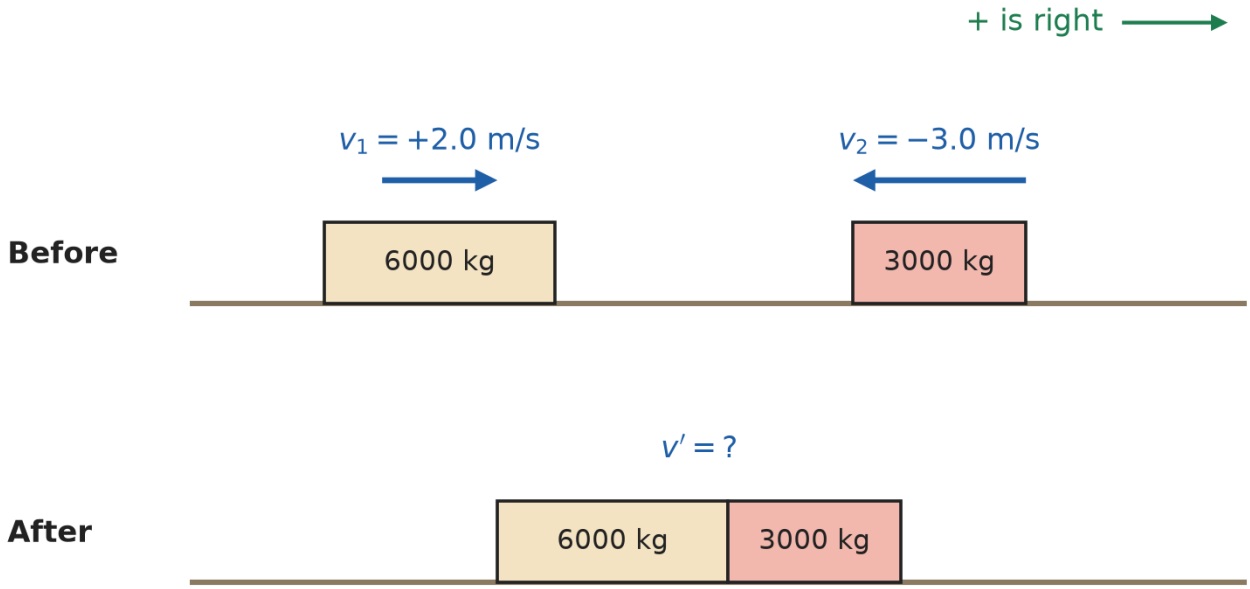
**GIVEN**

$m_1 = 6000$ kg  $v_1 = +2.0$ m/s  $m_2 = 3000$ kg

$v_2 = -3.0$ m/s  $v' = ?$

**Sign convention:** right is +, so "to the left at 3.0 m/s" is −3.0 m/s.

**Before and after (+ is right)**




The cars couple, so they share one final velocity v′. Isolate v′ in letters first.

Formula: $m_1v_1 + m_2v_2 = m_1v_1' + m_2v_2'$

Put in $v_1' = v_2' = v'$: $m_1v_1 + m_2v_2 = (m_1 + m_2)\,v'$

Divide both sides by $(m_1 + m_2)$: $v' = \dfrac{m_1v_1 + m_2v_2}{m_1 + m_2}$

Put in the numbers: $v' = \dfrac{(6000)(2.0) + (3000)(-3.0)}{6000 + 3000}$

$$v' = \frac{12000 - 9000}{9000} = +0.3333 \text{ m/s}$$

v′ is positive, so the pair rolls to the right.

**FINAL ANSWER**

**v′ = +0.33 m/s**

**$\vec{v}'$ = 0.33 m/s [right]**

QUESTION 2 [E]

## Recoil of a rifle

2. [E] Calculate the recoil velocity of an unconstrained rifle of mass 5.0 kg after it shoots a 50 g bullet at a speed of 300 m/s with respect to the earth.

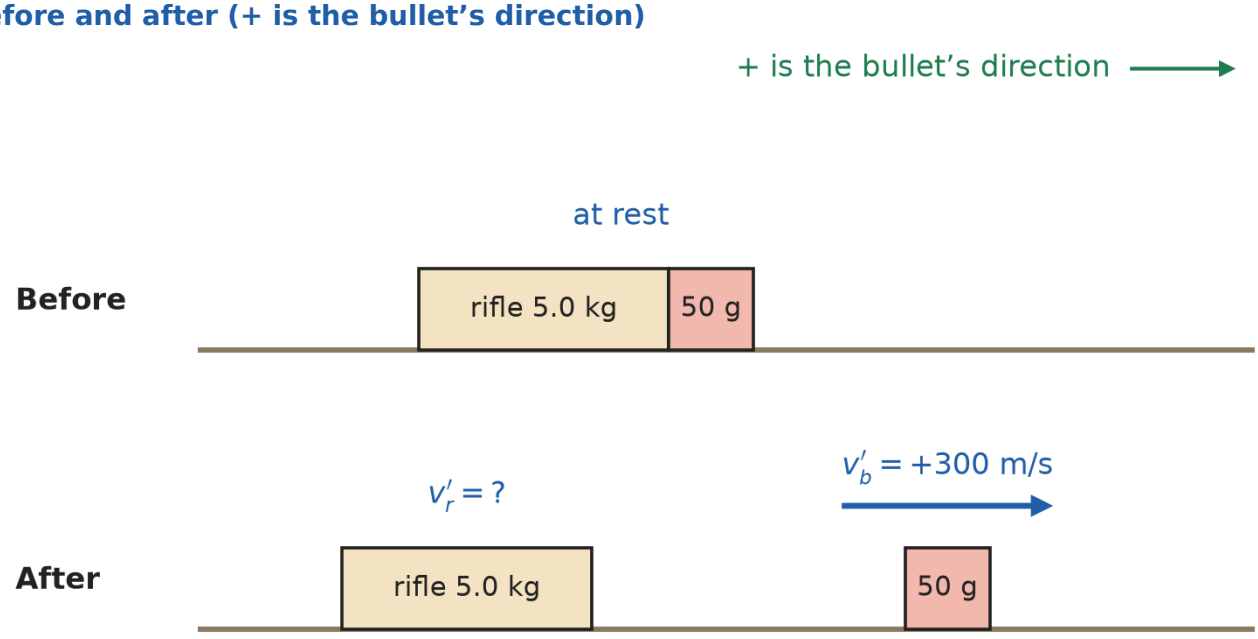
**GIVEN**

$m_r = 5.0$ kg $\qquad m_b = 50$ g $= 0.050$ kg

$v_b' = +300$ m/s $\qquad v_r' = ?$

**Sign convention:** the bullet's direction is +.

**Before and after (+ is the bullet's direction)**



Before the shot nothing is moving, so the total momentum is zero. It must still be zero after.

Formula: $m_r v_r + m_b v_b = m_r v_r' + m_b v_b'$

Put in $v_r = v_b = 0$: $0 = m_r v_r' + m_b v_b'$

Isolate $v_r'$: $v_r' = -\dfrac{m_b v_b'}{m_r}$

Put in the numbers: $v_r' = -\dfrac{(0.050)(300)}{5.0} = -3.00 \text{ m/s}$

The negative sign means the rifle moves opposite to the bullet.

**FINAL ANSWER**

**$v_r' = -3.0$ m/s**

**$\vec{v}_r' = 3.0$ m/s [opposite to the bullet]**

**QUESTION 3 [E]**

## Ball hits a stationary ball

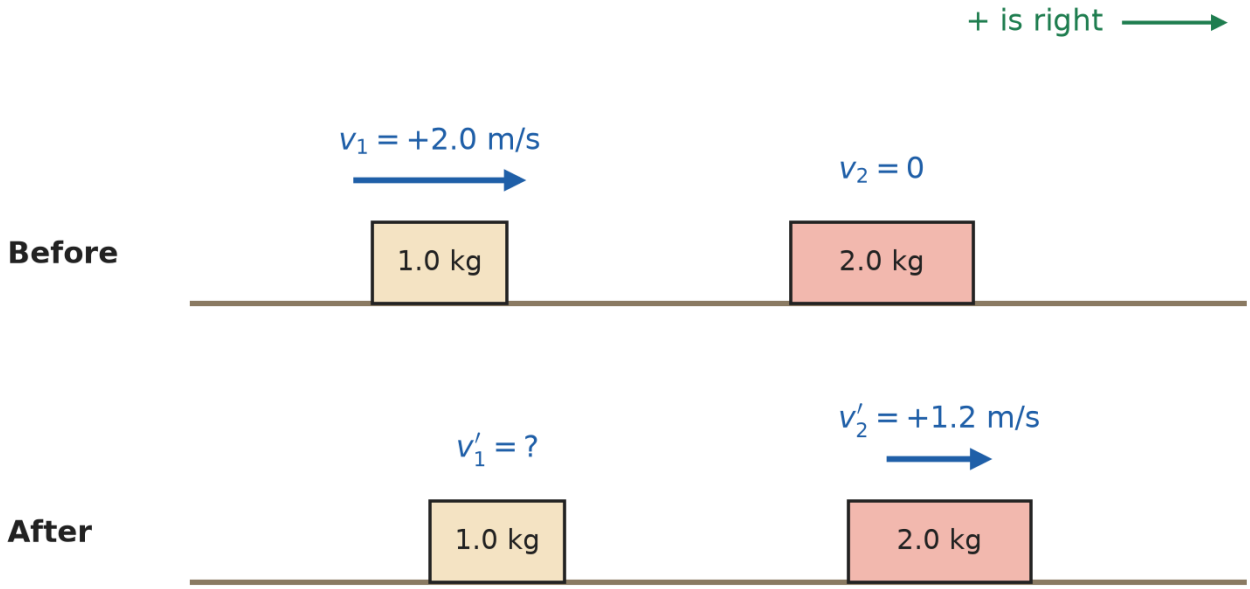
3. [E] A 1.0 kg ball moving with a velocity of 2.0 m/s to the right collides straight-on with a stationary 2.0 kg ball. After the collision, the 2.0 kg ball moves off to the right with a velocity of 1.2 m/s. What is the velocity of the 1.0 kg ball after the collision?

**GIVEN**

| | | |
|---|---|---|
| $m_1 = 1.0$ kg | $v_1 = +2.0$ m/s | $m_2 = 2.0$ kg |
| $v_2 = 0$ | $v_2' = +1.2$ m/s | $v_1' = ?$ |

**Sign convention:** right is +.

**Before and after (+ is right)**




Isolate $v_1'$ in letters first.

Formula: $m_1v_1 + m_2v_2 = m_1v_1' + m_2v_2'$

Subtract $m_2v_2'$, divide by $m_1$: $v_1' = \dfrac{m_1v_1 + m_2v_2 - m_2v_2'}{m_1}$

Put in the numbers:

$$v_1' = \frac{(1.0)(2.0) + (2.0)(0) - (2.0)(1.2)}{1.0} = \frac{2.0 - 2.4}{1.0} = -0.4 \text{ m/s}$$

The negative sign means the 1.0 kg ball bounces back to the left.

**FINAL ANSWER**

**$v_1' = -0.4$ m/s**

**$\vec{v}_1' = 0.4$ m/s [left]**

QUESTION 4 [M]

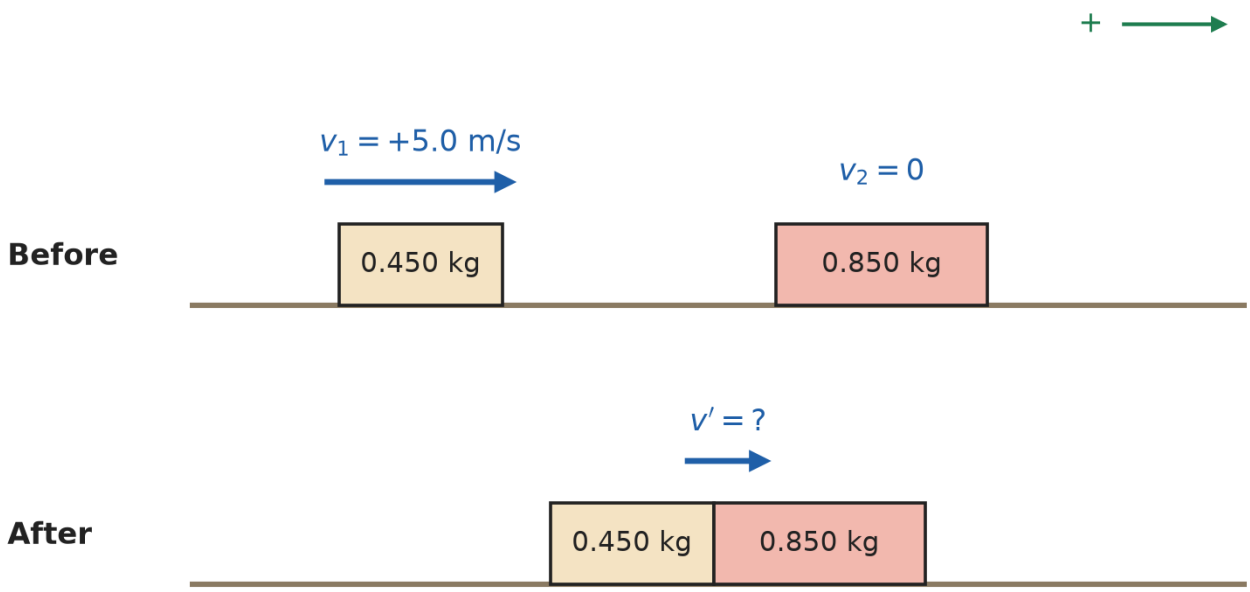
# Toy car, roller skate and spring

4. [M] A toy car (m = 0.450 kg) travelling at 5.0 m/s hits a roller skate with a lump of clay on it (m = 0.850 kg) and sticks to it. The two objects then travel together and encounter a spring (k = 50 N/m). How far will the spring compress?

**GIVEN**

| | | |
|---|---|---|
| $m_1 = 0.450$ kg | $v_1 = +5.0$ m/s | $m_2 = 0.850$ kg |
| $v_2 = 0$ | $k = 50$ N/m | $x = ?$ |

**The collision: before and after**




There are two stages. In the sticky collision momentum is conserved but kinetic energy is not. After the collision, while the spring is being compressed, energy is conserved.

## Step 1 – The collision (conservation of momentum)

The two objects stick, so they share one velocity v′.

Formula: $m_1 v_1 + m_2 v_2 = (m_1 + m_2)\, v'$

Isolate $v'$ (with $v_2 = 0$): $v' = \dfrac{m_1 v_1}{m_1 + m_2}$

Put in the numbers: $v' = \dfrac{(0.450)(5.0)}{0.450 + 0.850} = \dfrac{2.25}{1.300} = +1.7308$ m/s

### Step 2 – Compressing the spring (conservation of energy)

Start: the joined objects are moving, all kinetic. End: at rest at maximum compression, all elastic.

Formula: $E_{TOTi} = E_{TOTf}$

Put in the energies: $\frac{1}{2}(m_1 + m_2)v'^2 = \frac{1}{2}kx^2$

Multiply by $\frac{2}{k}$, square root: $x = v'\sqrt{\frac{m_1 + m_2}{k}}$

Put in the numbers: $x = (1.7308)\sqrt{\frac{1.300}{50}} = 0.2791 \text{ m}$

**FINAL ANSWER**

**x = 0.28 m**

**QUESTION 5 [H]**

## Elastic collision at the bottom of a track

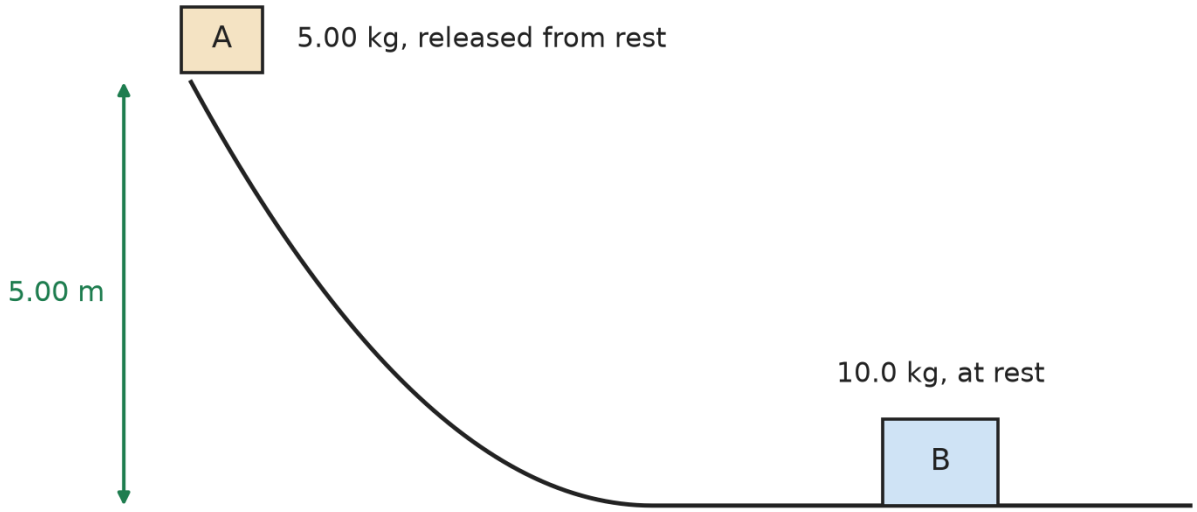
5. [H] Consider the frictionless track as shown below. A block of mass 5.00 kg is released from A. It makes a head-on elastic collision with a block of mass 10.0 kg at B which is initially at rest. Calculate the height to which the first block will rise after the collision.

**GIVEN**

| | | |
|---|---|---|
| $m_1 = 5.00$ kg | $m_2 = 10.0$ kg | $h = 5.00$ m |
| $v_2 = 0$ | $h' = ?$ | |

**Sign convention:** + is the direction block 1 moves along the level part (toward B).

**The track (h = 0 on the level part)**




frictionless track; h = 0 on the level part

### Step 1 – Speed of block 1 at the bottom (conservation of energy)

The track is frictionless, so all of the gravitational energy at A becomes kinetic energy at the bottom.

Formula: $E_{TOTi} = E_{TOTf} \quad \Rightarrow \quad m_1 gh = \frac{1}{2} m_1 v_1^2$

Isolate $v_1$: $v_1 = \sqrt{2gh} = \sqrt{2(9.81)(5.00)} = +9.9045$ m/s

### Step 2 – Velocity of block 1 after the elastic collision

Use the lesson's equation for a head-on elastic collision. Block 2 starts at rest, so the $v_2$ term is zero.

Formula: $v_1' = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) v_1 + \left(\frac{2m_2}{m_1 + m_2}\right) v_2$

Put in the numbers:

$$v_1' = \left(\frac{5.00 - 10.0}{5.00 + 10.0}\right)(9.9045) + 0 = -3.3015 \text{ m/s}$$

The negative sign means block 1 bounces back toward the curved part of the track.

### Step 3 – Height block 1 rises to (conservation of energy)

Going back up the track, its kinetic energy becomes gravitational energy again.

Formula: $\frac{1}{2}m_1 v_1'^{\,2} = m_1 g h'$

Isolate $h'$: $h' = \frac{v_1'^{\,2}}{2g}$

Put in the numbers: $h' = \frac{(-3.3015)^2}{2(9.81)} = 0.5556 \text{ m}$

Block 1 keeps one third of its speed, so it rises to one ninth of its starting height.

**FINAL ANSWER**

**h′ = 0.556 m**

**QUESTION 6 [H]**

## Two teams jump off a cart

6. [H] A competition is held between two teams of physics students, each team made up of 3 members and each member having a mass of 60 kg. The teams will take turns climbing onto a cart and jumping off. They want to see whose cart will obtain the greatest velocity after being propelled by a jump. They will start at the west end of a cart of mass 120 kg that is free to roll without friction on a level, east-west track. The plan is to run east along the cart and then jump off with a velocity if 10 m/s, with respect to the cart. The first team decides that its three members will run and jump off together. The second team decides that one member will depart according to the rules, followed by the second and then the third. Calculate the final velocity of the cart for each team and discuss the results.

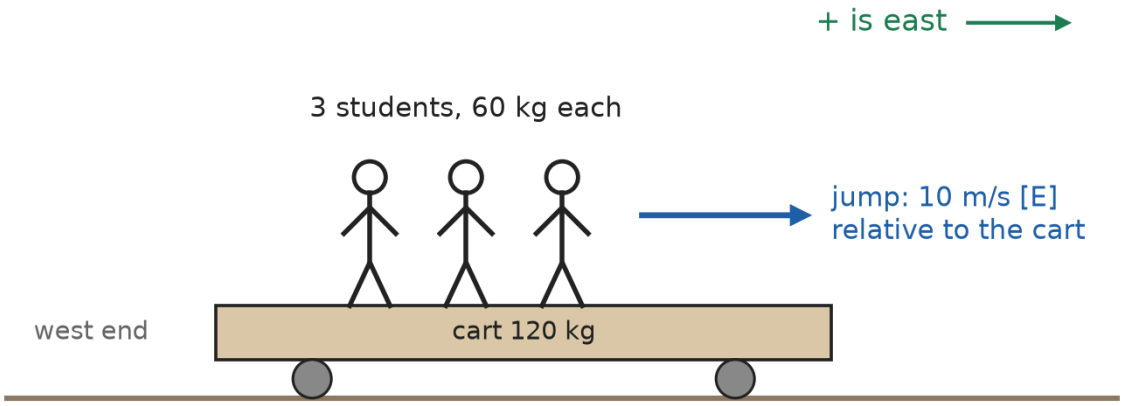
**GIVEN**

$m_{cart} = 120 \text{ kg}$ $\quad$ $m = 60 \text{ kg each}$

$v_{jc} = +10 \text{ m/s}$ $\quad$ $v'_{cart} = ?$

**Sign convention:** east is +.

### The cart and one team at the start




Momentum must be written with velocities measured relative to the ground. A jumper leaves at +10 m/s relative to the cart, and the cart is then moving at $v_c'$, so the jumper's velocity relative to the ground is $v_c' + 10$. The track is frictionless, so the total momentum of the cart and students is conserved in every jump.

**Team 1** its three members will run and jump off together

Before: everything is at rest, so the total momentum is zero. After: the cart (120 kg) moves at $v_c'$ and the three jumpers (180 kg) move at $v_c' + 10$.

Formula: $p_{tot} = p'_{tot}$

Put in the momenta: $0 = (120)\,v_c' + (180)(v_c' + 10)$

Expand and collect: $0 = 300\,v_c' + 1800$

Isolate $v_c'$: $v_c' = \dfrac{-1800}{300} = -6.0$ m/s

**Team 2** one member will depart according to the rules, followed by the second and then the third

### Jump 1

Before: at rest. After: the cart and two riders (240 kg) move at $v_A$, the jumper at $v_A + 10$.

Put in the momenta: $0 = (240)\,v_A + (60)(v_A + 10)$

Isolate $v_A$: $300\,v_A = -600 \quad \Rightarrow \quad v_A = -2.0$ m/s

### Jump 2

Before: the cart and two riders (240 kg) move at −2.0 m/s. After: the cart and one rider (180 kg) move at $v_B$, the jumper at $v_B$ + 10.

$$\text{Put in the momenta:}\quad (240)(-2.0) = (180)\,v_B + (60)(v_B + 10)$$

$$\text{Isolate } v_B\text{:}\quad 240\,v_B = -480 - 600 \quad \Rightarrow \quad v_B = -4.5 \text{ m/s}$$

### Jump 3

Before: the cart and one rider (180 kg) move at −4.5 m/s. After: the empty cart (120 kg) moves at $v_c'$, the jumper at $v_c'$ + 10.

$$\text{Put in the momenta:}\quad (180)(-4.5) = (120)\,v_c' + (60)(v_c' + 10)$$

$$\text{Isolate } v_c'\text{:}\quad 180\,v_c' = -810 - 600 \quad \Rightarrow \quad v_c' = -7.833 \text{ m/s}$$

**Discussion:** Team 2's cart ends up faster. Each jump speeds the cart up by more than the one before (2.0, then 2.5, then 3.3 m/s), because each jumper pushes off against a lighter cart-plus-riders. When all three jump together, they share a single push against the 120 kg cart.

**FINAL ANSWER**

**Team 1: $v_c'$ = −6.0 m/s; $\vec{v}_c'$ = 6.0 m/s [W]**

**Team 2: $v_c'$ = −7.8 m/s; $\vec{v}_c'$ = 7.8 m/s [W]**

**Jumping one at a time gives the cart the greater velocity**

QUESTION 7 [M-H]

# Girl walking on a plank

7. [M-H] A 45.0 kg girl is standing on a plank that has a mass of 150 kg. The plank, originally at rest, is free to slide on a frozen lake, which is a flat, frictionless supporting surface. The girl begins to walk along the plank at a constant speed of 1.5 m/s relative to the plank. (a) What is her speed relative to the ice surface? (b) What is the speed of the plank relative to the ice surface?

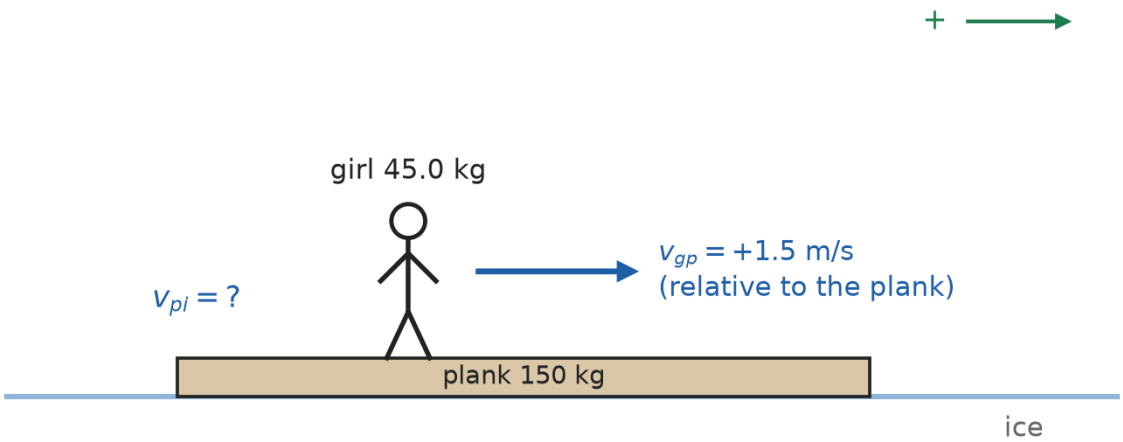
## GIVEN

$m_g = 45.0$ kg $\quad m_p = 150$ kg $\quad v_{gp} = +1.5$ m/s

$v_{gi} = ?$ $\quad v_{pi} = ?$

**Sign convention:** the direction the girl walks is +. Subscripts: gp = girl relative to plank, gi = girl relative to ice, pi = plank relative to ice.

### The girl and the plank (+ is the way she walks)




Momentum must use velocities relative to the ice. The girl's velocity relative to the ice is her velocity relative to the plank plus the plank's velocity relative to the ice. The plank's velocity is needed first, so part (b) is solved before part (a).

**(b)** What is the speed of the plank relative to the ice surface?

Everything starts at rest, so the total momentum is zero and stays zero.

Relative velocity: $v_{gi} = v_{gp} + v_{pi}$

Formula: $0 = m_g v_{gi} + m_p v_{pi}$

Put in $v_{gi}$: $0 = m_g(v_{gp} + v_{pi}) + m_p v_{pi}$

Isolate $v_{pi}$: $v_{pi} = -\dfrac{m_g v_{gp}}{m_g + m_p}$

Put in the numbers:

$$v_{pi} = -\frac{(45.0)(1.5)}{45.0 + 150} = -\frac{67.5}{195} = -0.34615 \text{ m/s}$$

The plank slides backward, opposite to the way she walks.

**(a)** What is her speed relative to the ice surface?

Relative velocity: $v_{gi} = v_{gp} + v_{pi}$

Put in the numbers: $v_{gi} = 1.5 + (-0.34615) = +1.15385$ m/s

**FINAL ANSWER**

**(a) Her speed relative to the ice = 1.2 m/s ($v_{gi}$ = +1.2 m/s)**

**(b) The plank's speed relative to the ice = 0.35 m/s ($v_{pi}$ = −0.35 m/s, opposite to her walking direction)**

**Worksheet answer:** (a) 1.15 m/s; (b) −0.346 m/s
**Why the answers differ:** Same values, different rounding. The key gets 1.1538 m/s and −0.3462 m/s. Her walking speed (1.5 m/s) has 2 significant figures, so the key gives 1.2 m/s and −0.35 m/s; the worksheet keeps 3.

QUESTION 8 [E]

## Golf ball

8. [E] A 0.046-kg golf ball is hit in such a way that its speed is 60 m/s immediately after being struck.

a. If the ball is in contact with the club head for 8.0 x $10^{-4}$ s, what is the average force on the ball?

b. If the angle of flight of the ball immediately after impact is $55^0$ to the horizontal, how far does the ball travel before landing on the ground? Assume no air resistance and point of impact for the ball is at same elevation as starting elevation.

**GIVEN**

$m = 0.046$ kg $\quad v_1 = 0 \quad v_2 = 60$ m/s

$\Delta t = 8.0 \times 10^{-4}$ s $\quad \theta = 55°$

**a.** If the ball is in contact with the club head for 8.0 x $10^{-4}$ s, what is the average force on the ball?

Impulse equals the change in momentum. The ball starts at rest. Take the direction it leaves as +.

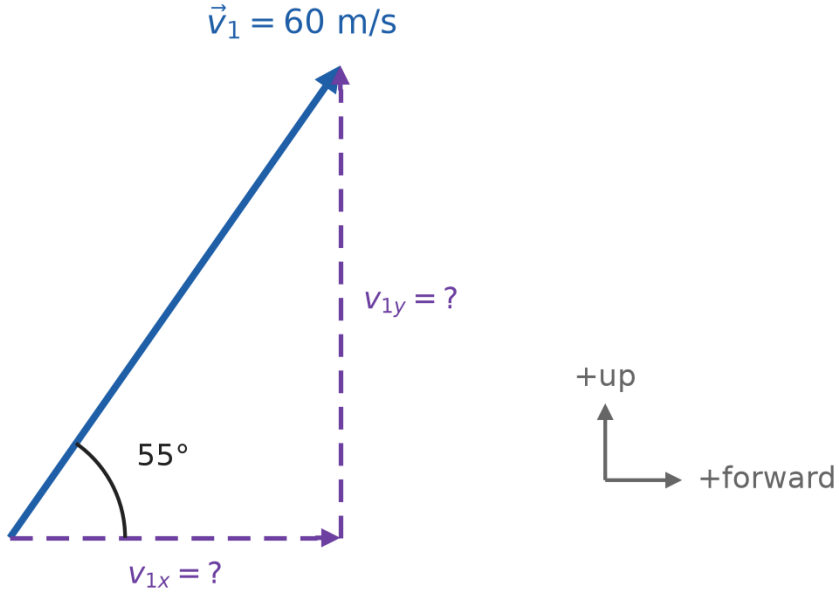
Formula: $F\Delta t = \Delta p = mv_2 - mv_1$

Divide both sides by $\Delta t$: $F = \dfrac{mv_2 - mv_1}{\Delta t}$

Put in the numbers: $F = \dfrac{(0.046)(60) - 0}{8.0 \times 10^{-4}} = 3450$ N

**b.** how far does the ball travel before landing on the ground?

**Launch velocity and its components (+x forward, +y up)**




### Step 1 – Components of the launch velocity

Horizontal: $v_{1x} = v_1 \cos\theta = (60)\cos 55° = 34.415$ m/s

Vertical: $v_{1y} = v_1 \sin\theta = (60)\sin 55° = 49.149$ m/s

### Step 2 – Time in the air (vertical motion)

It lands at the height it started from, so $\Delta d_y = 0$. With +y up, $a_y = -9.81$ m/s$^2$.

Formula: $\Delta d_y = v_{1y}\Delta t + \frac{1}{2}a_y \Delta t^2$

Put in $\Delta d_y = 0$ and divide by $\Delta t$: $0 = v_{1y} + \frac{1}{2}a_y \Delta t$

Isolate $\Delta t$: $\Delta t = \frac{-2v_{1y}}{a_y} = \frac{-2(49.149)}{-9.81} = 10.020$ s

### Step 3 – Horizontal distance

There is no horizontal acceleration.

Formula: $\Delta d_x = v_{1x}\,\Delta t$

Put in the numbers: $\Delta d_x = (34.415)(10.020) = 344.8$ m

**FINAL ANSWER**

**a. $F = 3.5 \times 10^3$ N**

**b. $\Delta d_x = 3.4 \times 10^2$ m**

QUESTION 9 [E]

## Model locomotive and car

9. [E] A 0.20-kg model railroad car moving at a speed of 0.24 m/s is struck from behind by a 0.42-kg model locomotive moving along the same line with a speed of 0.52 m/s. If they stick together after the collision, what is their velocity?

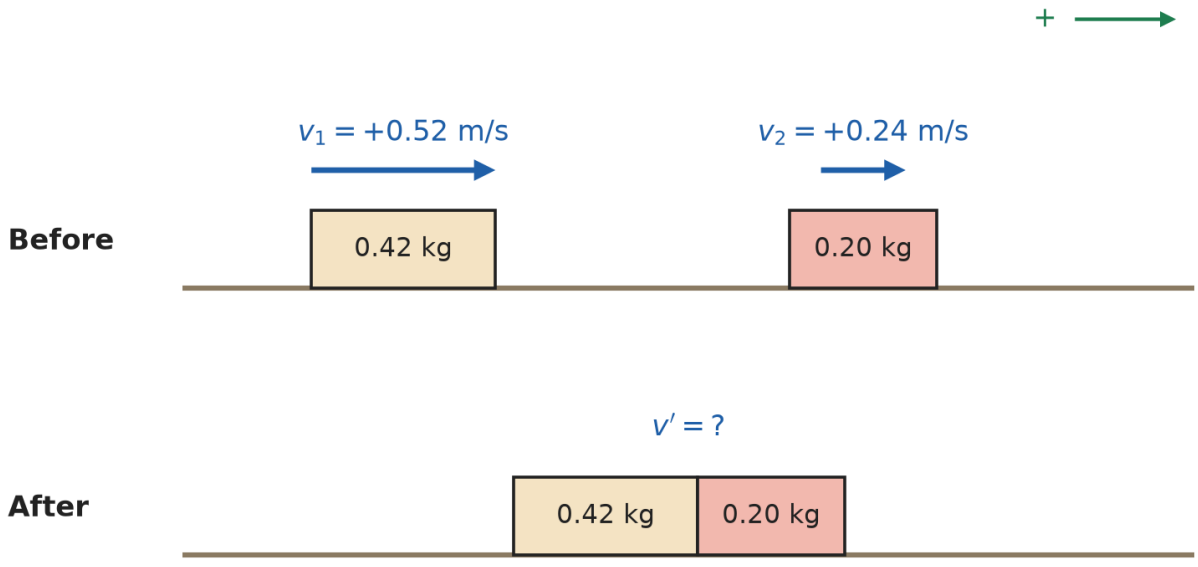
**GIVEN**

$m_1 = 0.42$ kg $\quad v_1 = +0.52$ m/s $\quad m_2 = 0.20$ kg

$v_2 = +0.24$ m/s $\quad v' = ?$

**Sign convention:** the direction both are moving is +.

**Before and after**



They stick together, so they share one final velocity v′.

Formula: $m_1v_1 + m_2v_2 = (m_1 + m_2)\,v'$

Divide both sides by $(m_1 + m_2)$: $v' = \dfrac{m_1v_1 + m_2v_2}{m_1 + m_2}$

Put in the numbers: $v' = \dfrac{(0.42)(0.52) + (0.20)(0.24)}{0.42 + 0.20}$

$$v' = \frac{0.2184 + 0.0480}{0.62} = +0.4297 \text{ m/s}$$

**FINAL ANSWER**

**v′ = +0.43 m/s**

**$\vec{v}'$ = 0.43 m/s [in the original direction of motion]**

**QUESTION 10 [E]**

## Two blocks and a spring

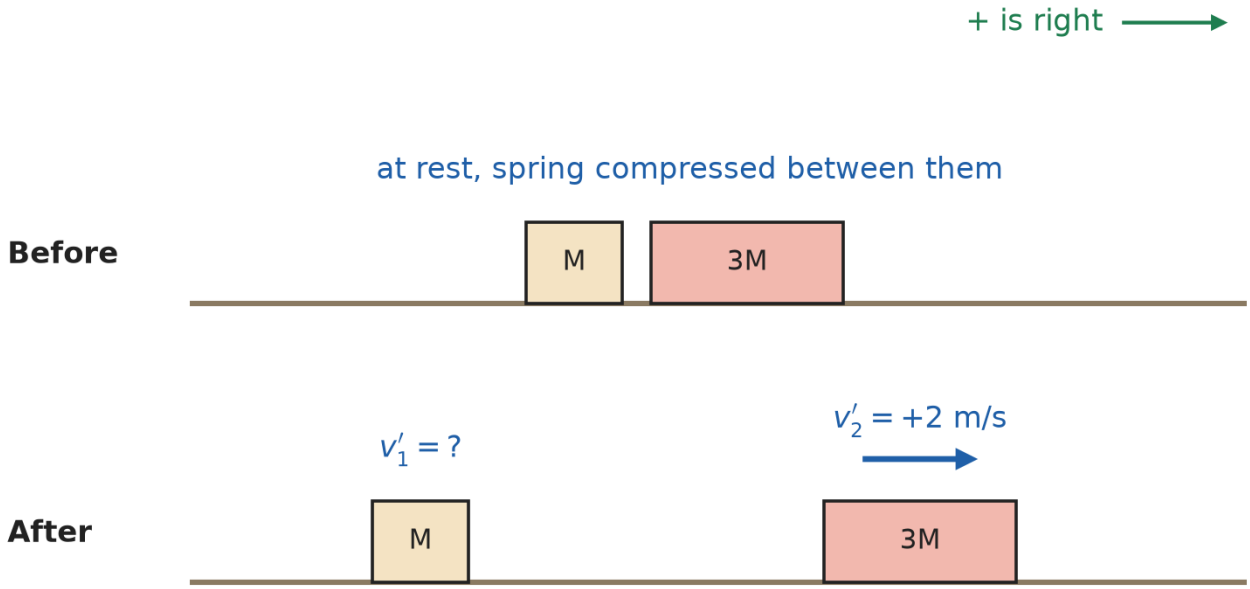
10. [E] Two blocks of mass M and 3M are placed on a horizontal, frictionless surface. A light spring is attached to one of them, and the blocks are pushed together with the spring between them. A string holding them together is burned, after which the block of mass 3M moves to the right with a speed of 2 m/s. What is the speed of mass M. (Assume that they are initially at rest)

**GIVEN**

$m_1 = M$ $m_2 = 3M$ $v_2' = +2$ m/s $v_1' = ?$

**Sign convention:** right is +.

**Before and after (+ is right)**




Both blocks start at rest, so the total momentum is zero before and after. M cancels.

Formula: $0 = m_1 v_1' + m_2 v_2'$

Put in the masses: $0 = Mv_1' + 3Mv_2'$

Divide by $M$, isolate $v_1'$: $v_1' = -3v_2'$

Put in the numbers: $v_1' = -3(2) = -6$ m/s

The negative sign means M moves to the left, three times as fast as 3M.

**FINAL ANSWER**

**$v_1' = -6$ m/s**

**The speed of mass M is 6 m/s; $\vec{v}_1' = 6$ m/s [left]**

QUESTION 11 [E]

# Car hits a truck from behind

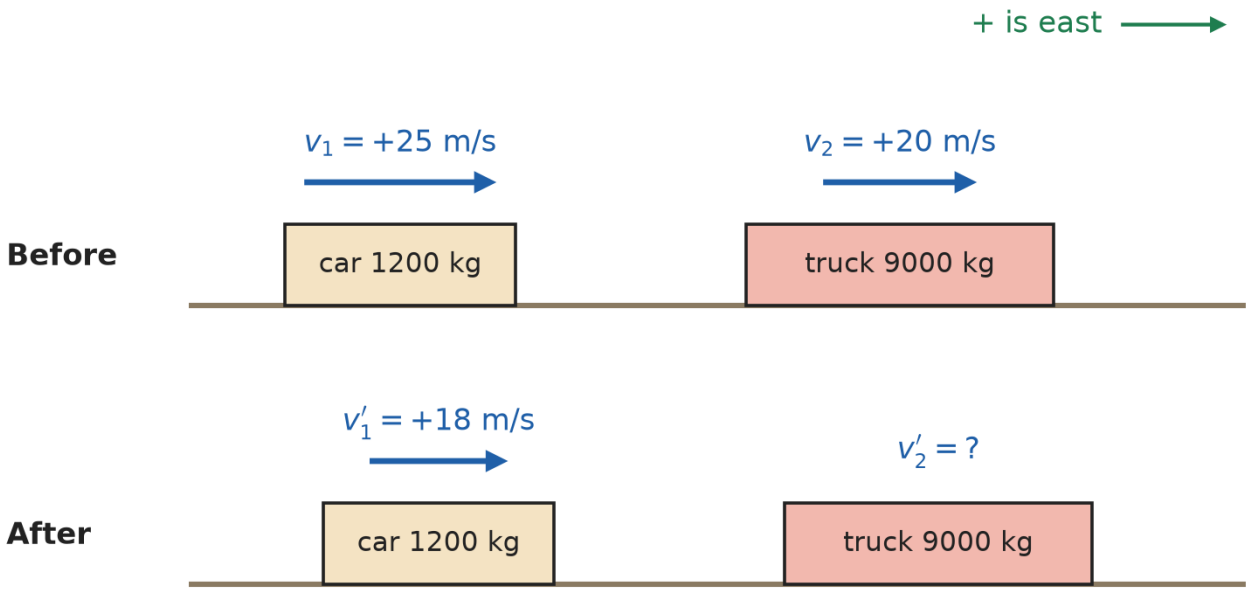
11. [E] A 1200-kg car travelling initially with a speed of 25 m/s in an easterly direction crashes into the rear end of a 9000-kg truck moving in the same direction at 20 m/s. The velocity of the car immediately after the collision is 18 m/s to the east. What is the velocity of the truck immediately after the collision?

## GIVEN

| | | |
|---|---|---|
| $m_1 = 1200$ kg | $v_1 = +25$ m/s | $v_1' = +18$ m/s |
| $m_2 = 9000$ kg | $v_2 = +20$ m/s | $v_2' = ?$ |

**Sign convention:** east is +.

## Before and after (+ is east)



Isolate $v_2'$ in letters first.

Formula: $m_1v_1 + m_2v_2 = m_1v_1' + m_2v_2'$

Subtract $m_1v_1'$, divide by $m_2$: $v_2' = \dfrac{m_1v_1 + m_2v_2 - m_1v_1'}{m_2}$

Put in the numbers: $v_2' = \dfrac{(1200)(25) + (9000)(20) - (1200)(18)}{9000}$

$$v_2' = \frac{30000 + 180000 - 21600}{9000} = +20.933 \text{ m/s}$$

**FINAL ANSWER**

**$v_2'$ = +21 m/s**

**$\vec{v}_2'$ = 21 m/s [E]**

**QUESTION 12 [M]**

## Two marbles, elastic collision

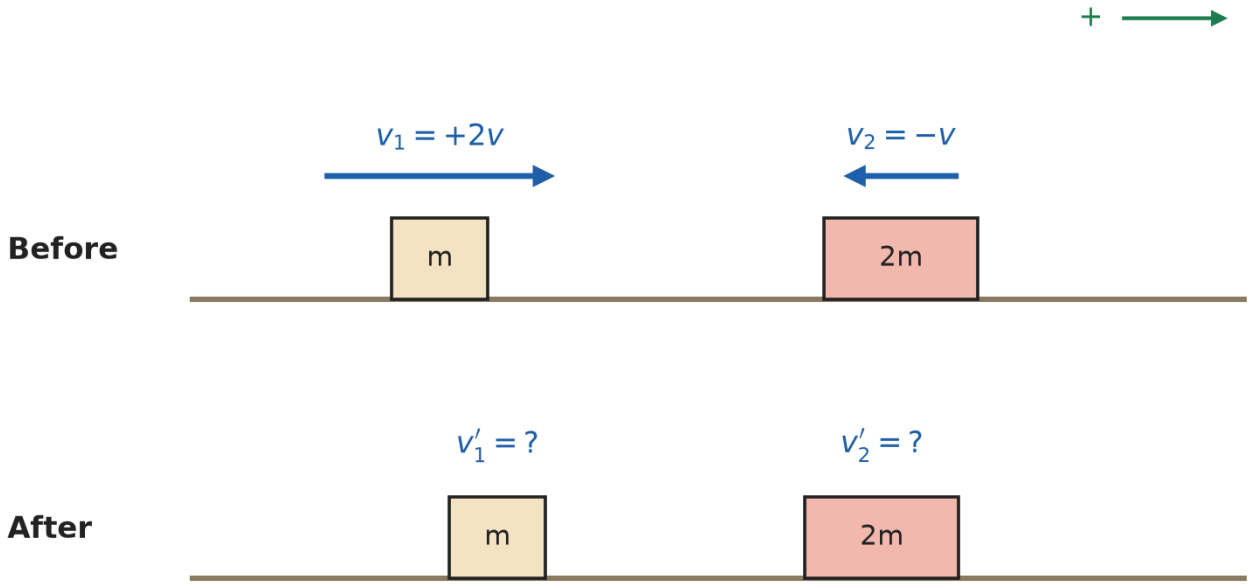
12. [M] Two glass marbles moving along a straight line towards each other undergo an elastic collision. The speed of one marble is 2v and its mass is m; the speed of the other marble is v and the mass is 2m. What are the speeds and directions of the two marbles after the collision?

**GIVEN**

| | | |
|---|---|---|
| $m_1 = m$ | $v_1 = +2v$ | $m_2 = 2m$ |
| $v_2 = -v$ | $v_1' = ?$ | $v_2' = ?$ |

**Sign convention:** the direction marble 1 (mass m) moves at first is +.

**Before and after**




## Step 1 – Marble 1

Use the lesson's equation for a head-on elastic collision. m cancels in each bracket.

Formula: $$v_1' = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)v_1 + \left(\frac{2m_2}{m_1 + m_2}\right)v_2$$

Put in the values: $$v_1' = \left(\frac{m - 2m}{m + 2m}\right)(2v) + \left(\frac{2(2m)}{m + 2m}\right)(-v)$$

Simplify: $$v_1' = \left(-\frac{1}{3}\right)(2v) + \left(\frac{4}{3}\right)(-v) = -\frac{2}{3}v - \frac{4}{3}v = -2v$$

### Step 2 – Marble 2

Formula: $v_2' = \left(\frac{m_2 - m_1}{m_1 + m_2}\right)v_2 + \left(\frac{2m_1}{m_1 + m_2}\right)v_1$

Put in the values: $v_2' = \left(\frac{2m - m}{m + 2m}\right)(-v) + \left(\frac{2m}{m + 2m}\right)(2v)$

Simplify: $v_2' = -\frac{1}{3}v + \frac{4}{3}v = +v$

Each marble reverses direction and keeps its speed. That happens because the total momentum before the collision is zero: m(2v) + 2m(−v) = 0.

**FINAL ANSWER**

**Marble of mass m: $v_1' = -2v$ (speed 2v, direction reversed)**

**Marble of mass 2m: $v_2' = +v$ (speed v, direction reversed)**

**So $v_1' = -2v_2'$**

**2 DIMENSIONS, QUESTION 1 [M]**

## Two cars at right angles

**Directions not given:** the question does not say which way each car is moving. The key takes the 1300 kg car moving up the page and the 1800 kg car moving right, which is the choice that matches the direction in the worksheet's answer.

1. [M] Two cars approaching each other at right angles collide and stick together. One car has a mass of 1300 kg and an initial speed of 2.25 m/s while the other car has a mass of 1800 kg and an initial speed of 4.50 m/s. What will be their speed and direction immediately after impact?

**GIVEN**

$m_1 = 1300$ kg

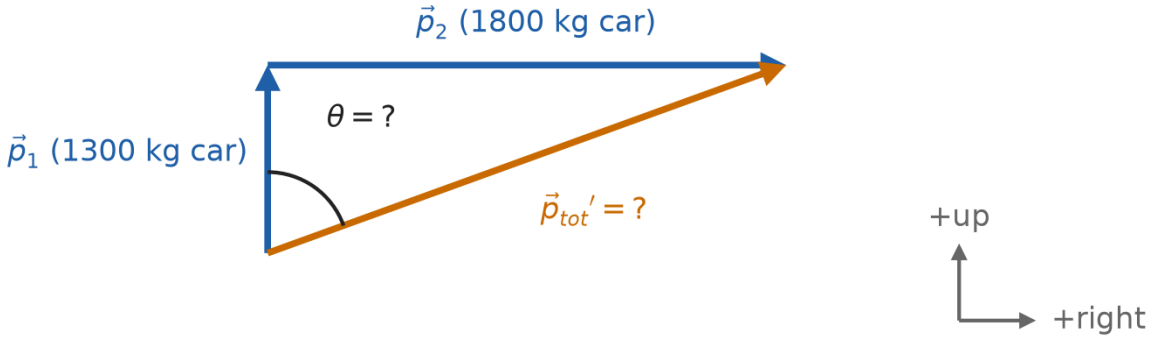
$v_1 = 2.25$ m/s [up]

$m_2 = 1800$ kg

$v_2 = 4.50$ m/s [right]

$\vec{v}' = ?$

**The two momentum vectors, added tip to tail**




### Step 1 – Momentum in each direction

Car 1 has only y-momentum and car 2 has only x-momentum. The cars stick together, so the wreck carries all of both.

x (right): $p'_x = m_2 v_2 = (1800)(4.50) = 8100$ kg·m/s

y (up): $p'_y = m_1 v_1 = (1300)(2.25) = 2925$ kg·m/s

### Step 2 – Total momentum and speed

The two components are perpendicular, so use the Pythagorean theorem.

Size: $p'_{tot} = \sqrt{p'_x{}^2 + p'_y{}^2} = \sqrt{(8100)^2 + (2925)^2} = 8611.9$ kg·m/s

Speed: $v' = \dfrac{p'_{tot}}{m_1 + m_2} = \dfrac{8611.9}{3100} = 2.7780$ m/s

### Step 3 – Direction

θ is measured from “up” toward “right”, so the opposite side is the x-momentum.

Formula: $\tan\theta = \dfrac{p'_x}{p'_y}$

Put in the numbers: $\theta = \tan^{-1}\left(\dfrac{8100}{2925}\right) = 70.14°$

**FINAL ANSWER**

**$\vec{v}'$ = 2.8 m/s [up 70° right]**

2 DIMENSIONS, QUESTION 2

# Grenade in three pieces

2. A grenade at rest explodes into three different pieces. The first piece goes off with a velocity of 55 m/s [N] and the second piece moves with a velocity of 85 m/s [W]. Determine the velocity of the third piece of the grenade. The second piece has half the mass of the first piece and the third piece is three times the mass of the first piece.

## GIVEN

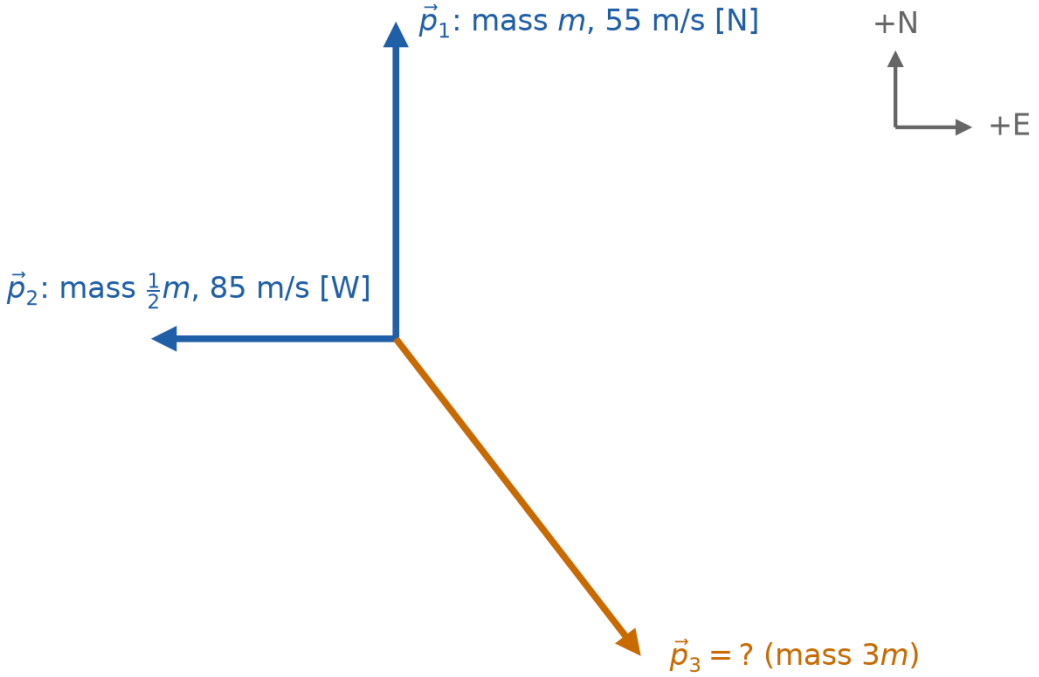
$m_1 = m$ $\qquad$ $\vec{v}_1{}' = 55$ m/s [N]

$m_2 = \frac{1}{2}m$ $\qquad$ $\vec{v}_2{}' = 85$ m/s [W]

$m_3 = 3m$ $\qquad$ $\vec{v}_3{}' = ?$

**Sign convention:** east is +x and north is +y, so 85 m/s [W] is −85 m/s in x.

## The three momentum vectors




The grenade starts at rest, so the three momenta must add to zero. Piece 3 must go south and east to cancel the other two.

### Step 1 – x-momentum (east–west)

Only pieces 2 and 3 move in x. m cancels.

Formula: $0 = m_2 v'_{2x} + m_3 v'_{3x}$

Put in the masses: $0 = \frac{1}{2} m(-85) + 3m\, v'_{3x}$

Divide by $m$, isolate $v'_{3x}$: $v'_{3x} = \frac{(0.5)(85)}{3} = +14.167$ m/s

### Step 2 – y-momentum (north–south)

Only pieces 1 and 3 move in y.

Formula: $0 = m_1 v'_{1y} + m_3 v'_{3y}$

Put in the masses: $0 = m(55) + 3m\, v'_{3y}$

Divide by $m$, isolate $v'_{3y}$: $v'_{3y} = -\frac{55}{3} = -18.333$ m/s

### Step 3 – Speed and direction

$v_{3x}'$ is east and $v_{3y}'$ is south. The angle is measured from south toward east.

Speed: $v'_3 = \sqrt{(14.167)^2 + (18.333)^2} = 23.169$ m/s

Direction: $\theta = \tan^{-1}\left(\frac{14.167}{18.333}\right) = 37.69°$

**FINAL ANSWER**

**$\vec{v}_3'$ = 23 m/s [S 38° E]**

2 DIMENSIONS, QUESTION 3 [H]

# The boy, the train and the flowers

3. [H] A railroad track lies alongside a frozen lake. A train moves along the track with a constant speed of 7.0 m/s. A boy on a frictionless ice sled is initially moving parallel to the track with the same speed as the train and 10 m away from it. The boy and sled together weigh 50 kg and the boy carries a 2.0-kg bouquet of flowers with him. At some time the boy tosses the flowers to a girl on the train with a velocity of 15.0 m/s perpendicular to the direction of his motion. The girl on the train catches the flowers and immediately throws them back to the boy, who catches it. The girl throws the bouquet perpendicular to her motion with the same speed at which she caught it. What is the final direction and speed of the boy and sled?

**GIVEN**

$m_b = 50$ kg (boy + sled)

$m_f = 2.0$ kg (flowers)

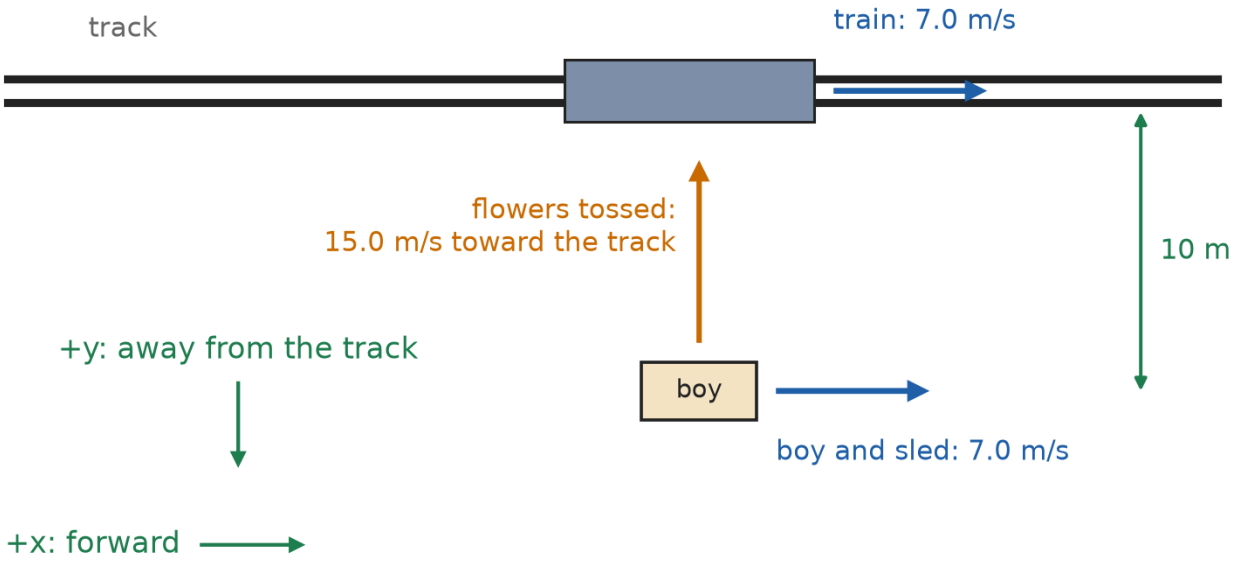
$v_x = 7.0$ m/s

$v_f = 15.0$ m/s

$\vec{v}_b{}' = ?$

**Sign convention:** +x is forward along the track and +y is away from the track.

**View from above, at the moment of the toss**



The boy, the flowers and the train all move forward at 7.0 m/s the whole time, and every throw is perpendicular to that, so the forward (x) velocities never change. Only the y-momentum is exchanged. The train is so massive that catching and throwing the flowers does not change its motion.

### Step 1 – The toss (y-momentum of boy + flowers)

Before the toss neither has any y-velocity. The flowers leave at 15.0 m/s toward the track, which is −15.0 m/s in y.

Formula: $0 = m_b v_{by} + m_f v_{fy}$

Isolate $v_{by}$: $v_{by} = -\dfrac{m_f v_{fy}}{m_b} = -\dfrac{(2.0)(-15.0)}{50} = +0.600 \text{ m/s}$

The boy now drifts away from the track at 0.600 m/s.

### Step 2 – The catch (y-momentum of boy + flowers)

The girl throws the flowers back at the same speed, now away from the track: +15.0 m/s in y. The boy catches them, so they share one y-velocity afterward.

Formula: $m_b v_{by} + m_f v_{fy} = (m_b + m_f)\, v'_y$

Isolate $v'_y$: $v'_y = \dfrac{m_b v_{by} + m_f v_{fy}}{m_b + m_f}$

Put in the numbers:

$$v'_y = \frac{(50)(0.600) + (2.0)(15.0)}{50 + 2.0} = \frac{60.0}{52} = +1.1538 \text{ m/s}$$

### Step 3 – Final speed and direction

The forward velocity is still 7.0 m/s.

$$\text{Speed:}\quad v' = \sqrt{v_x^2 + v_y'^2} = \sqrt{(7.0)^2 + (1.1538)^2} = 7.0945 \text{ m/s}$$

$$\text{Direction:}\quad \theta = \tan^{-1}\left(\frac{v_y'}{v_x}\right) = \tan^{-1}\left(\frac{1.1538}{7.0}\right) = 9.360°$$

**FINAL ANSWER**

**$\vec{v}_b' = 7.1$ m/s [forward 9.4° away from the track]**

**2 DIMENSIONS, QUESTION 4 [M]**

## Two pucks

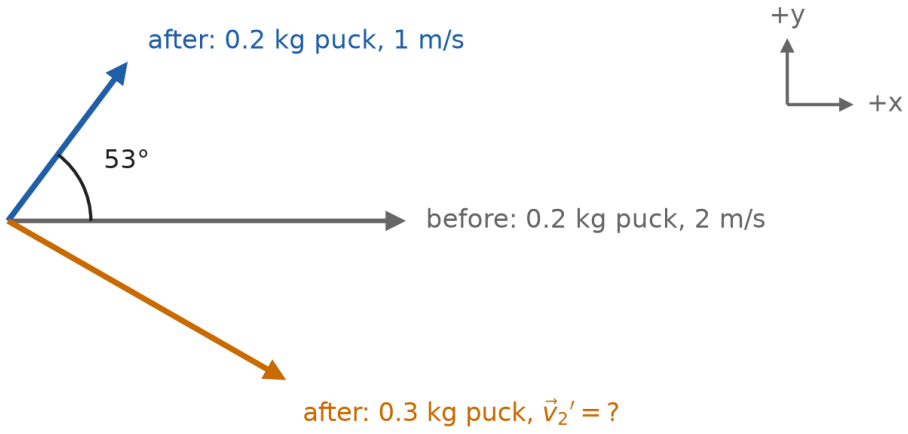
4. [M] A 0.3-kg puck, initially at rest on a horizontal, frictionless surface, is struck by a 0.2-kg puck moving initially along the x-axis with a velocity of 2 m/s. After the collision, the 0.2-kg puck has a speed of 1 m/s at an angle of θ = $53^0$ to the positive x-axis. Determine the velocity of the 0.3-kg puck after the collision.

**GIVEN**

| | | |
|---|---|---|
| $m_1 = 0.2$ kg | $v_1 = 2$ m/s [+x] | $v_1' = 1$ m/s, 53° |
| $m_2 = 0.3$ kg | $v_2 = 0$ | $\vec{v}_2' = ?$ |

**Sign convention:** +x is the first puck's original direction; +y is the side of the x-axis the 0.2 kg puck goes to.

**Velocities before and after**



### Step 1 – x-momentum

Before, only the 0.2 kg puck moves, along +x.

Formula: $m_1 v_1 = m_1 v_1' \cos 53° + m_2 v_{2x}'$

Isolate $v_{2x}'$: $v_{2x}' = \dfrac{m_1 v_1 - m_1 v_1' \cos 53°}{m_2}$

Put in the numbers:

$$v_{2x}' = \frac{(0.2)(2) - (0.2)(1)\cos 53°}{0.3} = \frac{0.4 - 0.1204}{0.3} = +0.9321 \text{ m/s}$$

### Step 2 – y-momentum

There is no y-momentum before the collision, so the total must stay zero.

Formula: $0 = m_1 v_1' \sin 53° + m_2 v_{2y}'$

Isolate $v_{2y}'$:

$$v_{2y}' = -\frac{m_1 v_1' \sin 53°}{m_2} = -\frac{(0.2)(1)\sin 53°}{0.3} = -0.5324 \text{ m/s}$$

The negative sign means the 0.3 kg puck goes to the other side of the x-axis.

### Step 3 – Speed and direction

Speed: $v_2' = \sqrt{(0.9321)^2 + (0.5324)^2} = 1.0735 \text{ m/s}$

Direction: $\theta = \tan^{-1}\left(\dfrac{0.5324}{0.9321}\right) = 29.73°$ below the $+x$ axis

**FINAL ANSWER**

**$\vec{v}_2'$ = 1 m/s [30° from the +x axis, on the opposite side from the 0.2 kg puck]**

**Worksheet answer:** 1 m/s [29° from horizontal]
**Why the answers differ:** Same speed; the angle is rounded differently.
$\tan^{-1}(0.5324/0.9321) = 29.73°$, which rounds to 30°. The worksheet cut it off at 29°.